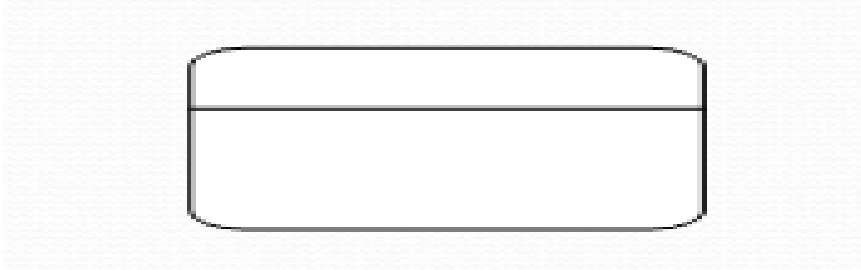
Data flow Diagram

Data Flow Diagram (DFD)

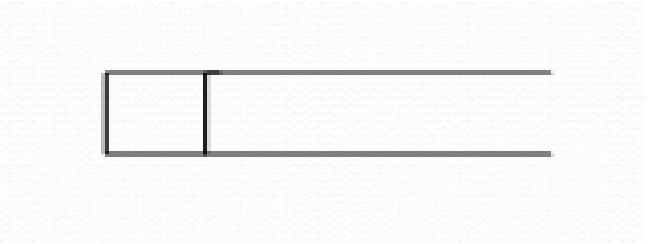
Data flow diagram (DFD) is a picture of the movement of data between external entities and the processes and data stores within a system

DFD Symbols

Process Notations. A process transforms incoming data flow into outgoing data flow.

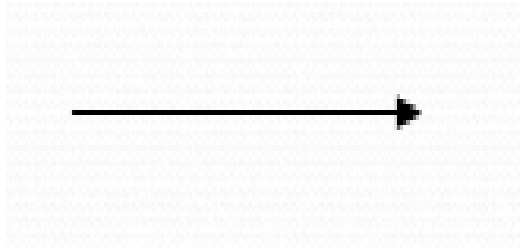


- **Datastore Notations.** Datastores are repositories of data in the system. They are sometimes also referred to as files

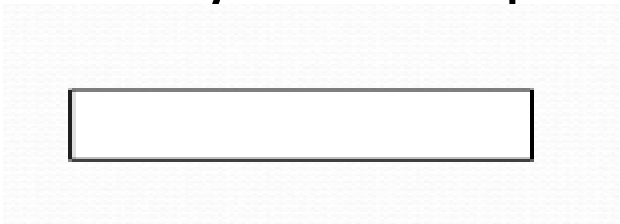


DFD Symbols

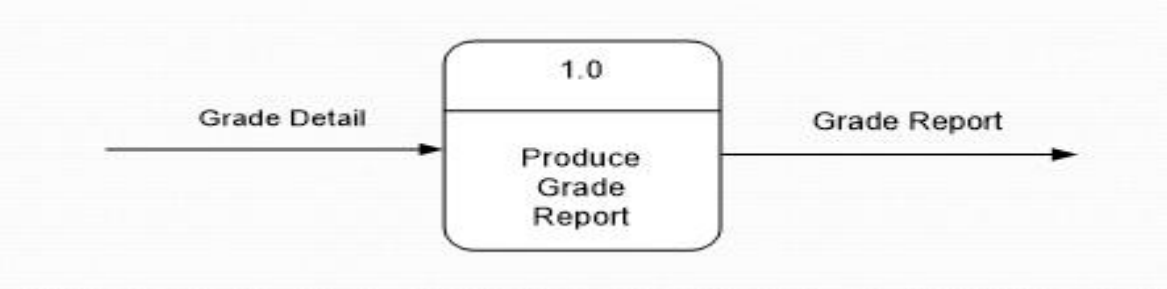
Dataflow Notations. Dataflows are pipelines through which packets of information flow. Label the arrows with the name of the data that moves through it



External Entity Notations. External entities are objects outside the system, with which the system communicates. External entities are sources and destinations of the system's inputs and outputs



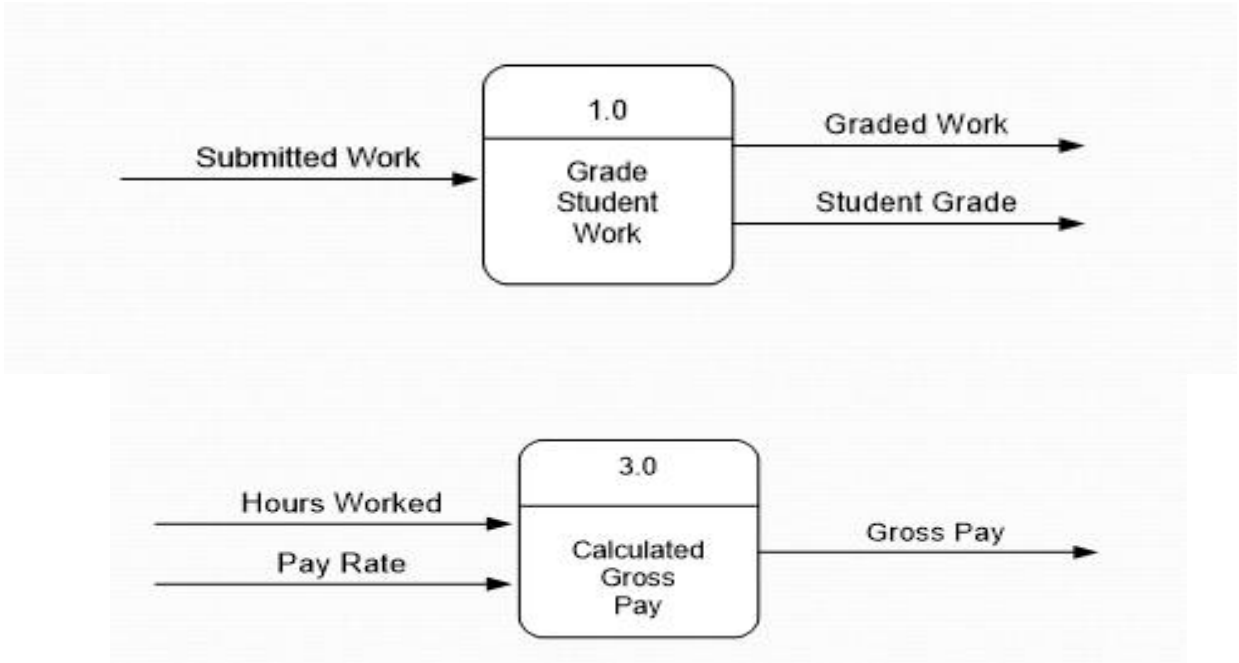
Process



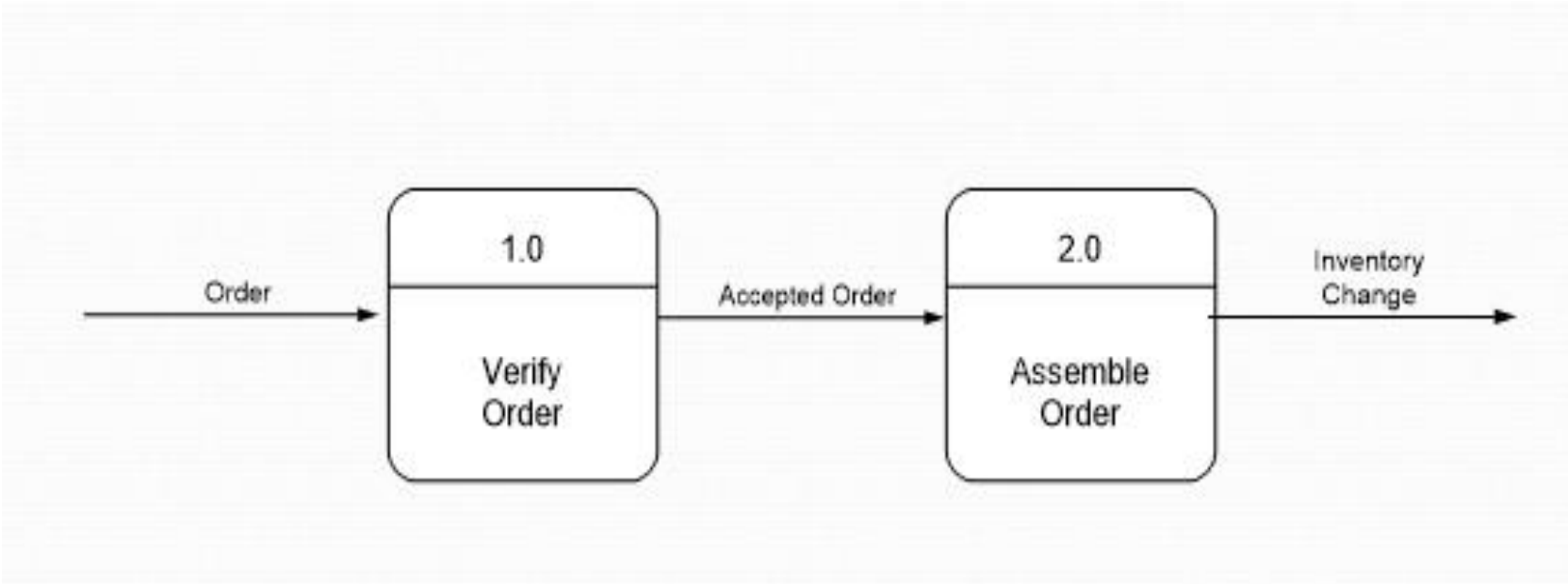
- Work or actions performed on data (inside the system)
- Labels should be **verb phrases**
- Receives input data and produces output
- Must be numbered

Rule 1: Process

- Can have more than one outgoing data flow or more than one incoming data flow



Rule 2: Process



Context Data flow (level 0 DFD)

- A context data flow diagram (DFD), also known as a level 0 DFD, gives a broad overview of an information system and the way it interacts with external entities.

Example1: Mail Ordering System

Consider a mail order company that distributes CDs and tapes at discount prices to its members.

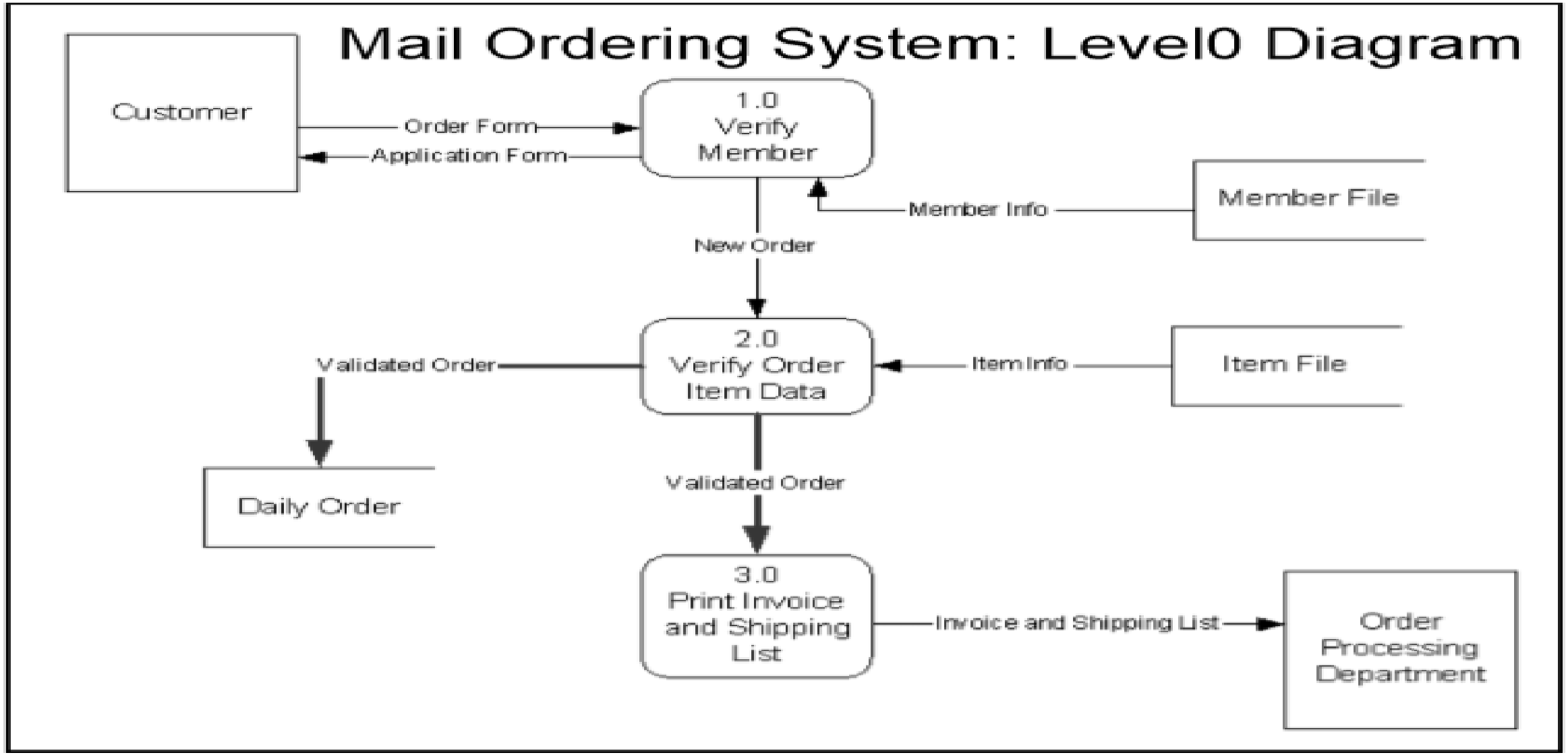
- When an order processing clerk receives an order form, she verifies that the sender is a member by checking the MEMBER FILE.
- If the sender is not a member, the clerk returns the order along with a membership application form.
- If the customer is a member, the clerk verifies the order item data by checking the ITEM FILE.
- Then the clerk enters the order data and saves it to the DAIILY ORDERS FILE.
- At the same time the clerk also prints an invoice and shipping list for each order, which are forwarded to the ORDER FULFILLMENT DEPARTMENT for processing there.

Example1: Mail Ordering System

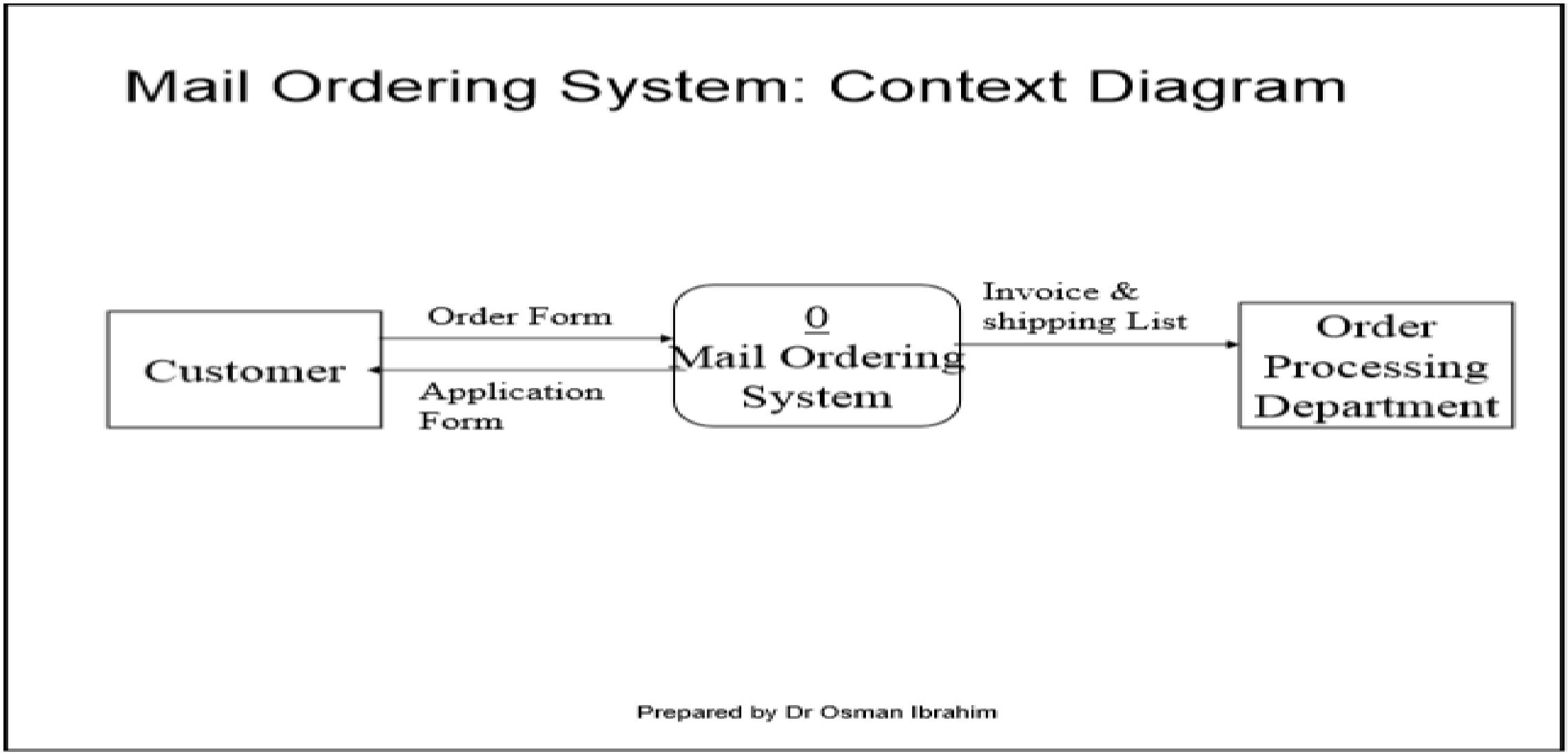
A mail order company that distributes CDs and tapes at discount prices to record club members (tells the purpose of the system).

- When an order processing clerk receives an order form (data flow), she verifies (process #1) that the sender is a club member by checking the MEMBER FILE (data store). If the sender is not a member, the clerk returns the order along with a membership application form (data flow).
- If the customer is a member, the clerk verifies the order item data (process #2) by checking the ITEM FILE (data store).
- Then the clerk enters the order data (data flow) and saves it to the DAILY ORDERS FILE (data store).
- At the same time the clerk also prints an invoice and shipping list (data flow) for each order (process #3), which are forwarded to the ORDER FULFILLMENT DEPARTMENT (external entity) for processing there.

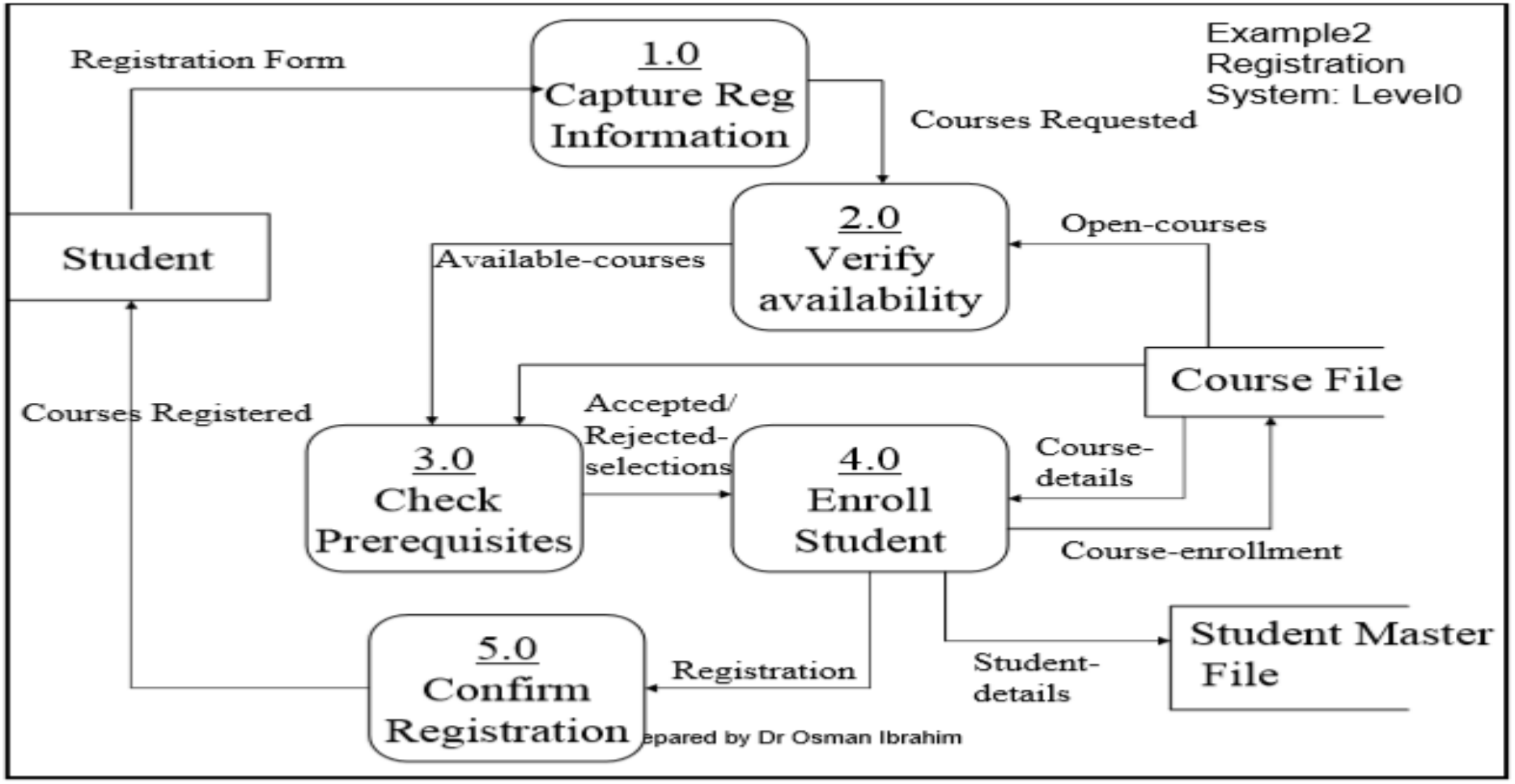
Mail ordering System Data flow



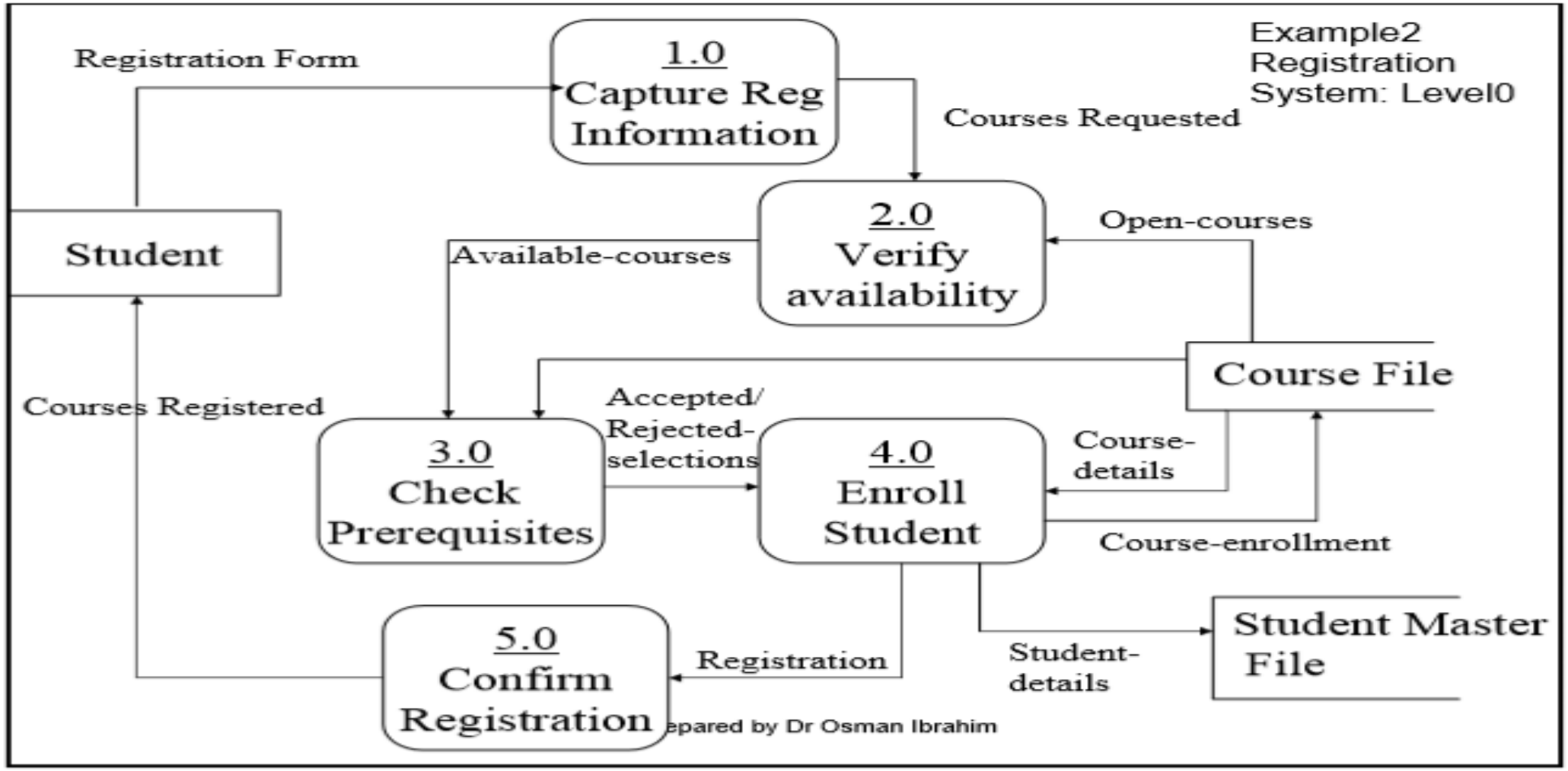
Mail ordering System Context Diagram



Course Registration Data flow



Course Registration Context Diagram



Thank you